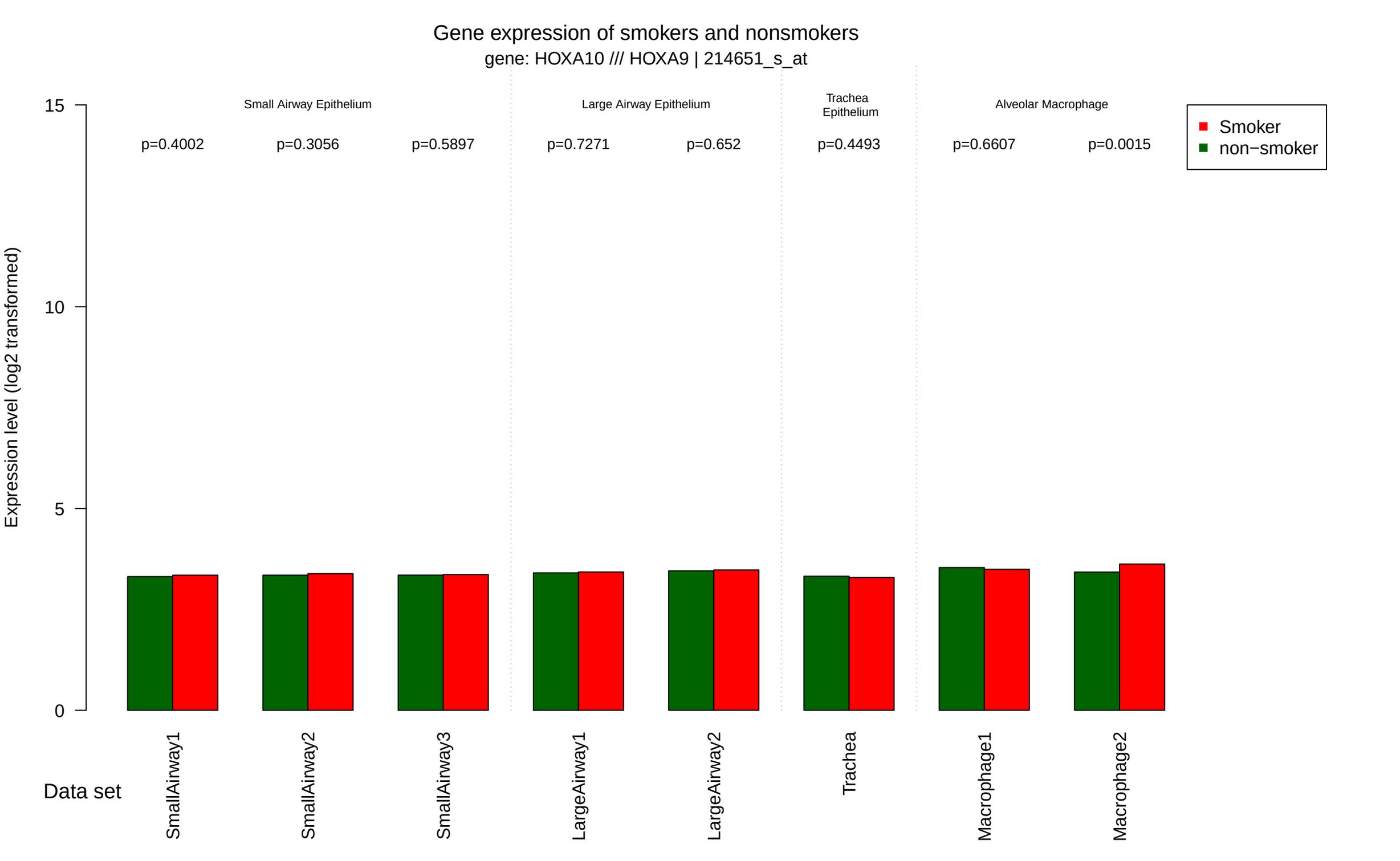
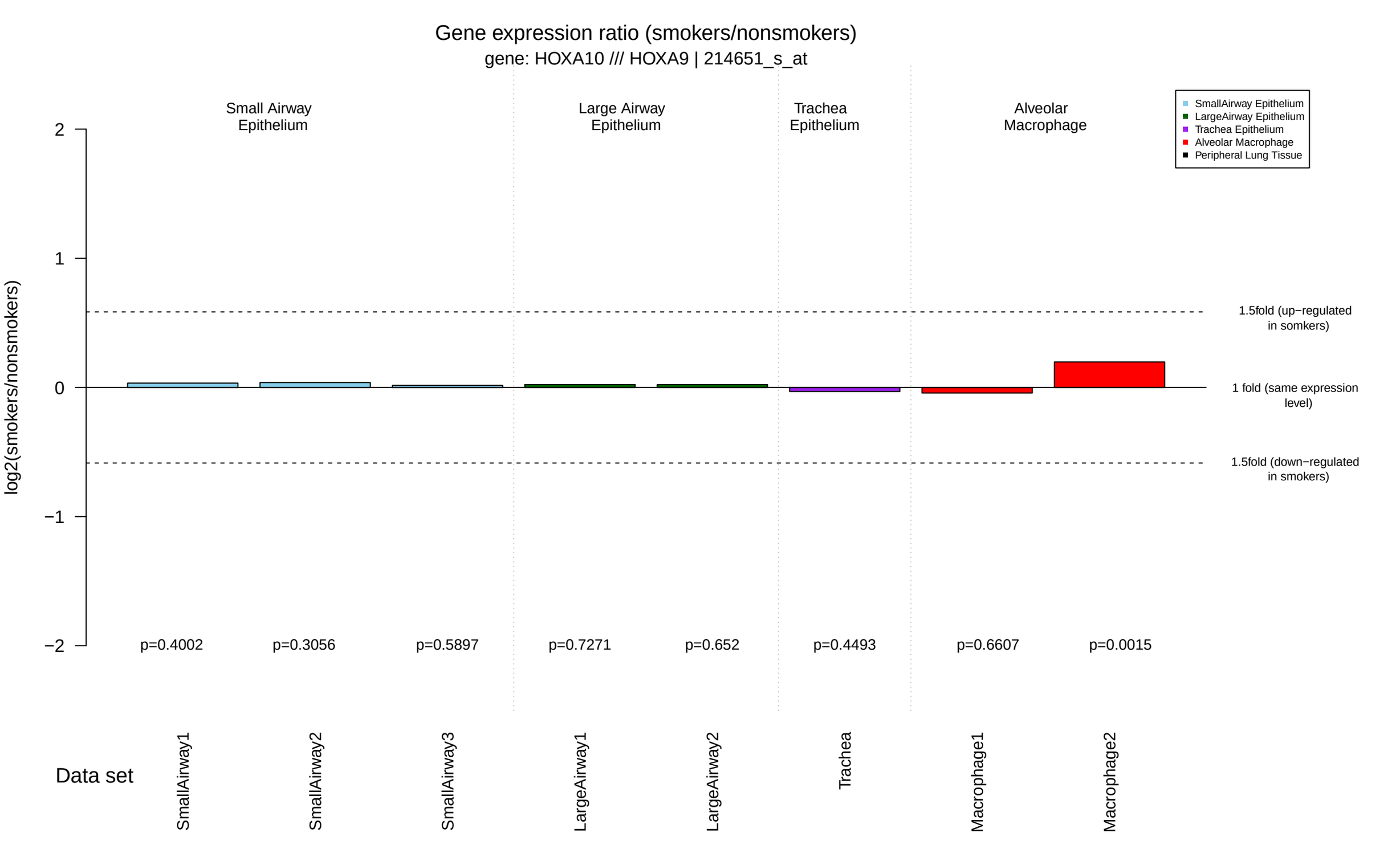
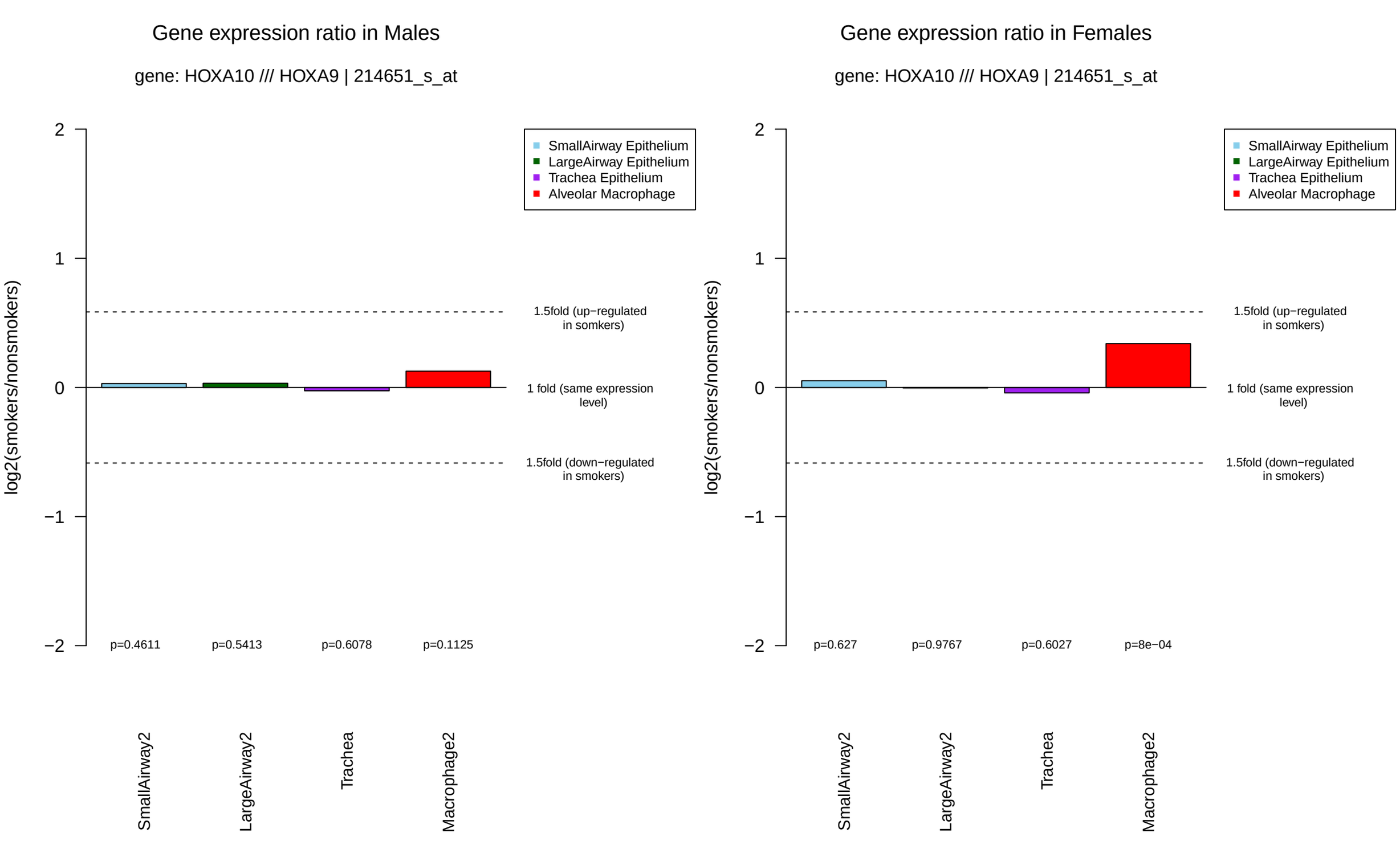
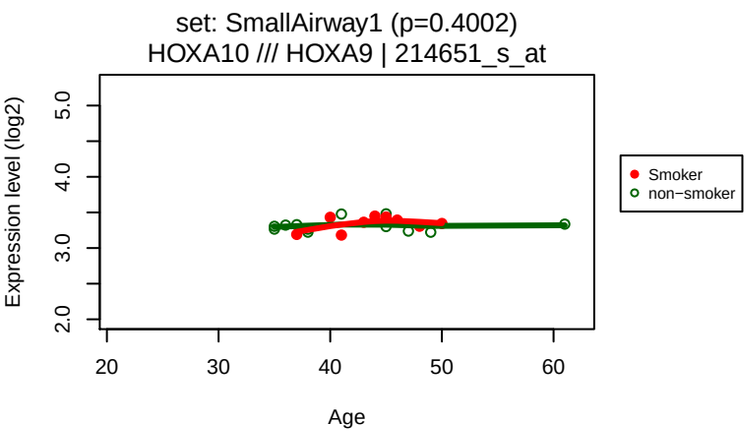
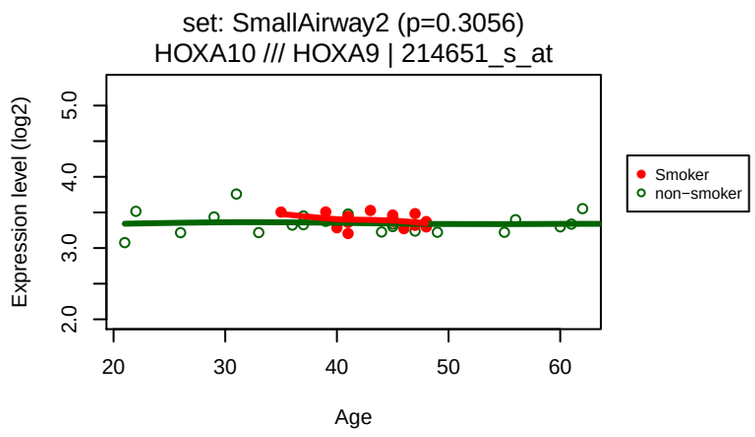
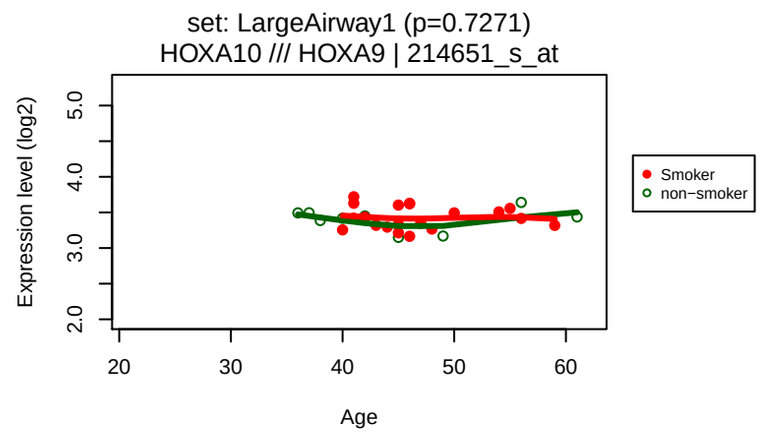

Gene expression of smokers and nonsmokers
gene: HOXA10 /// HOXA9 | 214651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4002
p=0.3056
p=0.5897
p=0.7271
p=0.652
p=0.4493
p=0.6607
p=0.0015
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

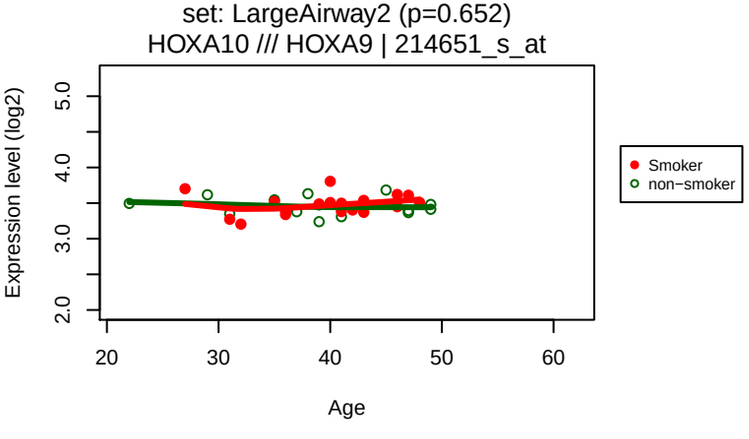

Gene expression ratio (smokers/nonsmokers)
gene: HOXA10 /// HOXA9 | 214651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4002
p=0.3056
p=0.5897
p=0.7271
p=0.652
p=0.4493
p=0.6607
p=0.0015
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

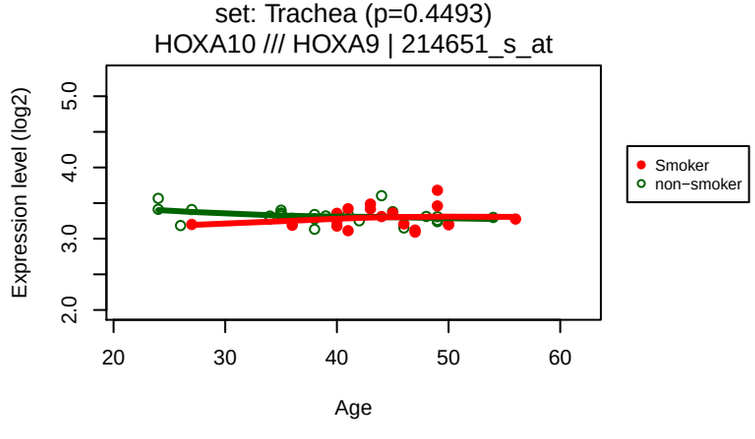

Gene expression ratio in Males
gene: HOXA10 /// HOXA9 | 214651_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4611
p=0.5413
p=0.6078
p=0.1125
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA10 /// HOXA9 | 214651_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.627
p=0.9767
p=0.6027
p=8e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2

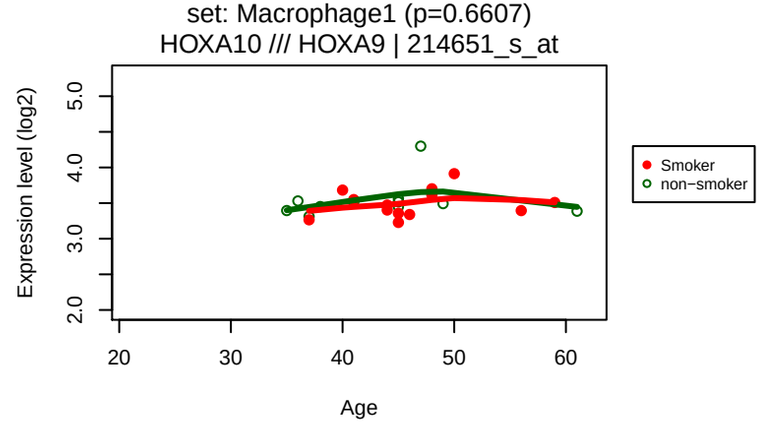

set: SmallAirway1 (p=0.4002)
HOXA10 /// HOXA9 | 214651_s_at
Expression level (log2)
Age
Smoker
non−smoker

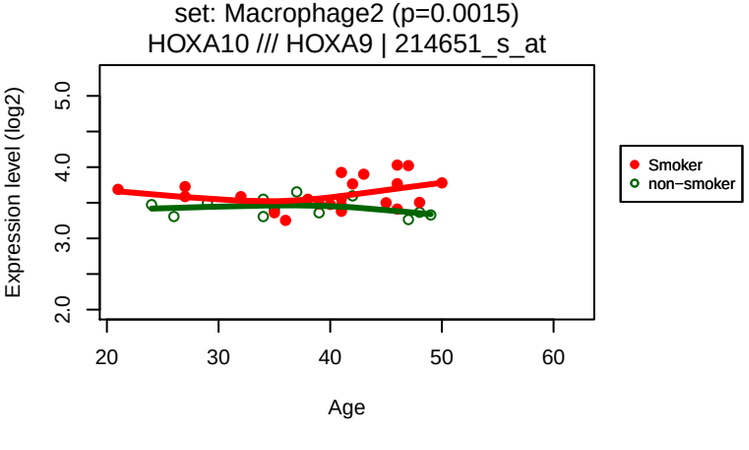

set: SmallAirway2 (p=0.3056)
HOXA10 /// HOXA9 | 214651_s_at
Expression level (log2)
Age
Smoker
non−smoker

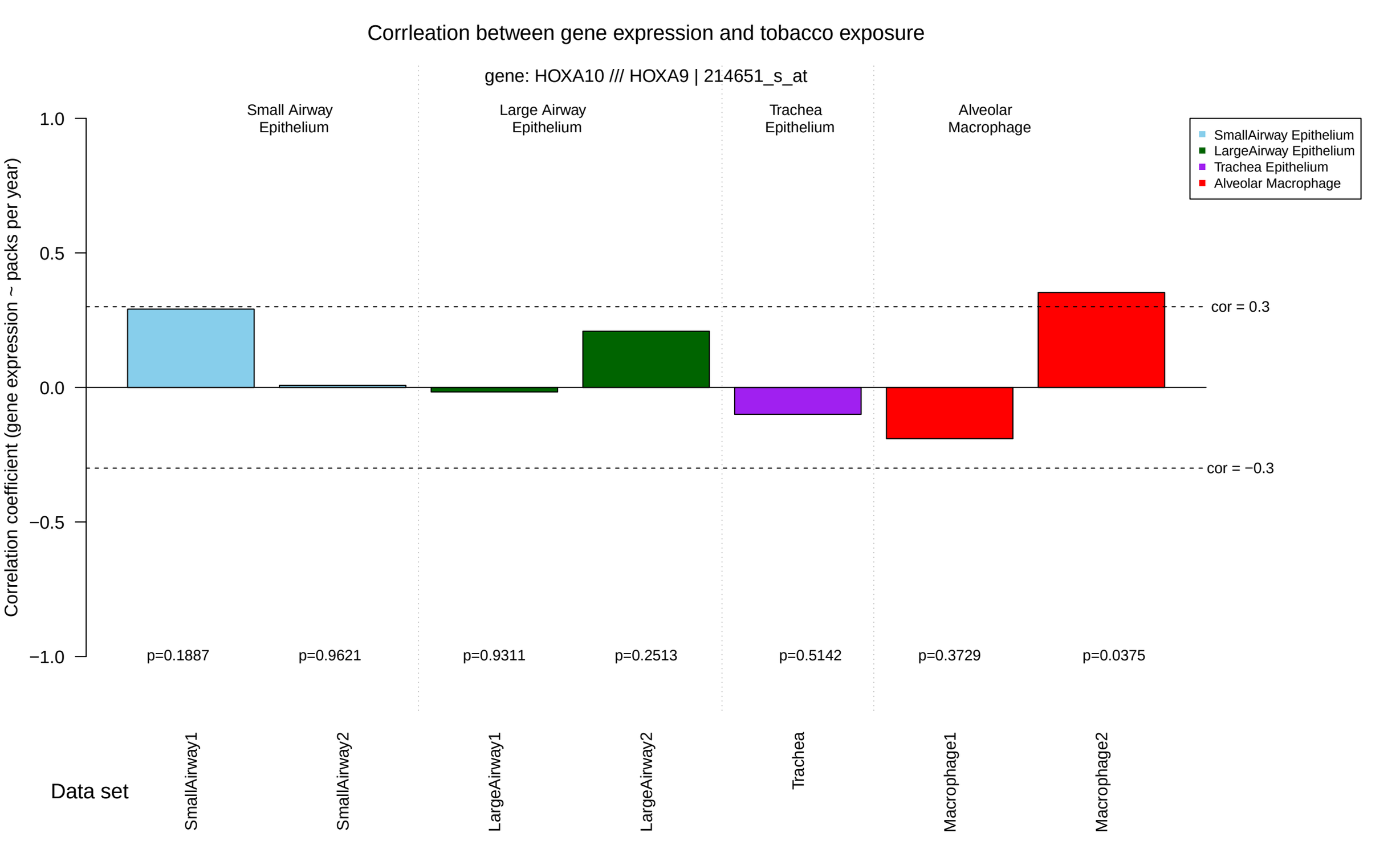

set: LargeAirway1 (p=0.7271)
HOXA10 /// HOXA9 | 214651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.652)
HOXA10 /// HOXA9 | 214651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4493)
HOXA10 /// HOXA9 | 214651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6607)
HOXA10 /// HOXA9 | 214651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0015)
HOXA10 /// HOXA9 | 214651_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA10 /// HOXA9 | 214651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1887
p=0.9621
p=0.9311
p=0.2513
p=0.5142
p=0.3729
p=0.0375
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2